

Teaching Note

CVP Analysis Microsimulation

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1. Platform Overview

1.1 Game Arena

The gaming arena interface comprises various decision tabs that includes market, decision areas such as Marketing, Production, Decision checklist, Report, and Synopsis. Some sections must be completed first as they have an impact on other areas.

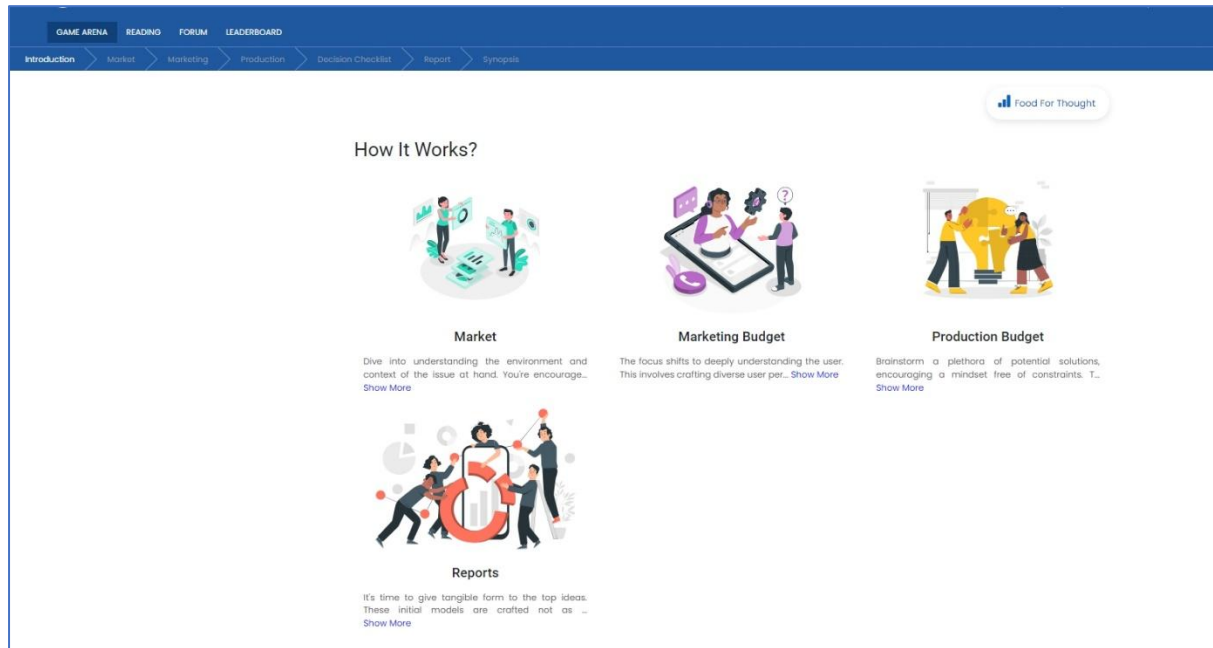


Figure 1 The Gaming Arena

1.2 Readings

This section provides essential materials necessary for participants to actively engage in the game. The main reading material consists of a guide for decision-making, supplemented by optional additional resources chosen by instructors. Moreover, a walkthrough video is accessible, offering in-depth information about the scenario featured in the simulation. This video presents insights into the case scenario, market outlook and the diverse array of decisions required.

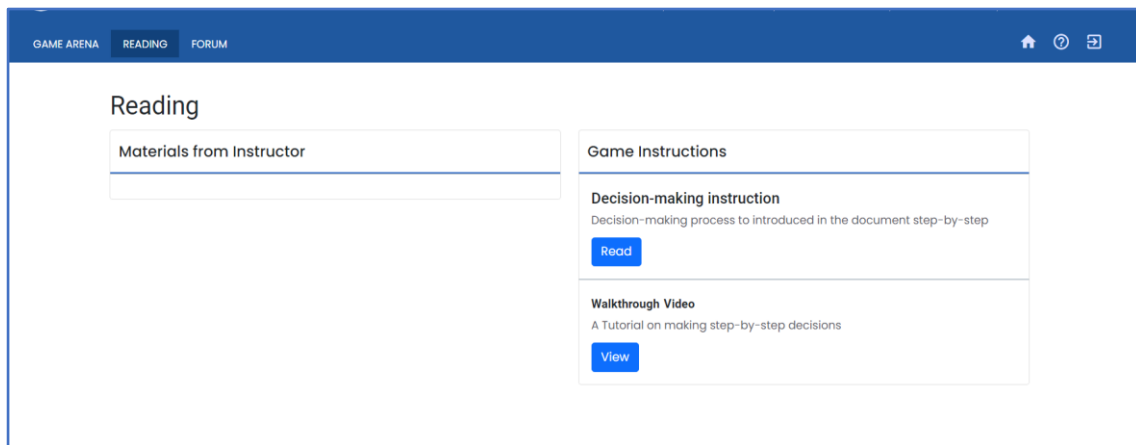


Figure 2 The Reading Section

1.3 Forum

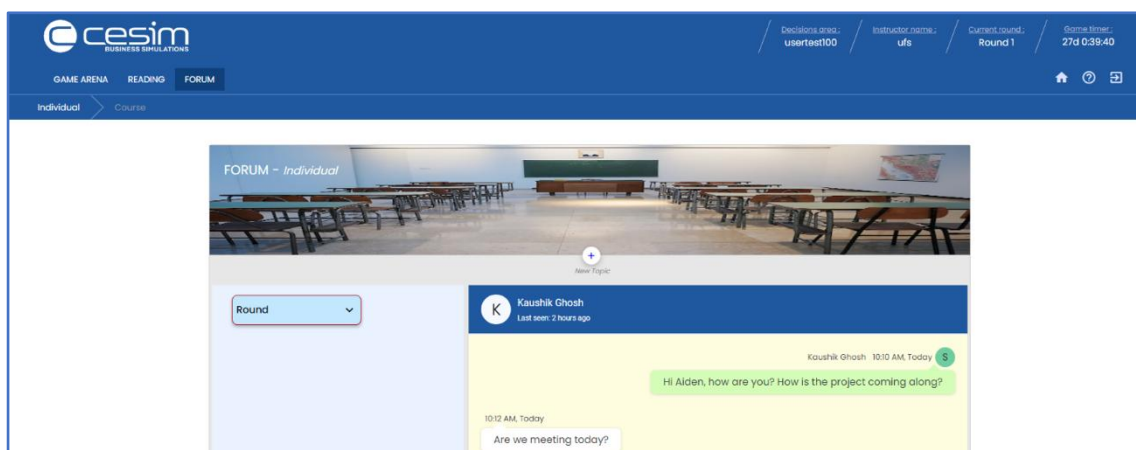


Figure 3 The Forum Section

- The forums provide an excellent platform for participants to engage with their instructors or peers.
- There are two separate forums: the Individual Forum and the Course Forum. The Individual Forum allows participants to communicate directly with the instructor, while the Course Forum is accessible to all enrolled players.
- Instructors can view and respond to posts in both sections. Consequently, the Course Forum is an appropriate space for asking questions that can benefit the entire course, while the Individual Forum is the preferred channel for the individual to have a direct discussion with the professor.
- Participants will receive notifications for messages.

1.4 Main Objective & Winning Criteria

A leading clothing manufacturer specializing in high-quality Jeans and Tops holds a 30% market share in a local market that is growing at 5%. With a strong emphasis on competitive pricing and product quality, the company is well-positioned to meet the rising demand. The industry is highly competitive, each excelling in pricing, quality, and innovation. As the factory manager, participants are tasked with optimizing profit margins through strategic decisions on pricing, quality, and design, leveraging insights into competitors' strengths and weaknesses. Success of the simulation lies in optimizing production efficiency, achieving the KPI, of a better operating margin of 10% or higher.

1.5 Successful decision-making flow

The sequence of decision-making unfolds as follows:

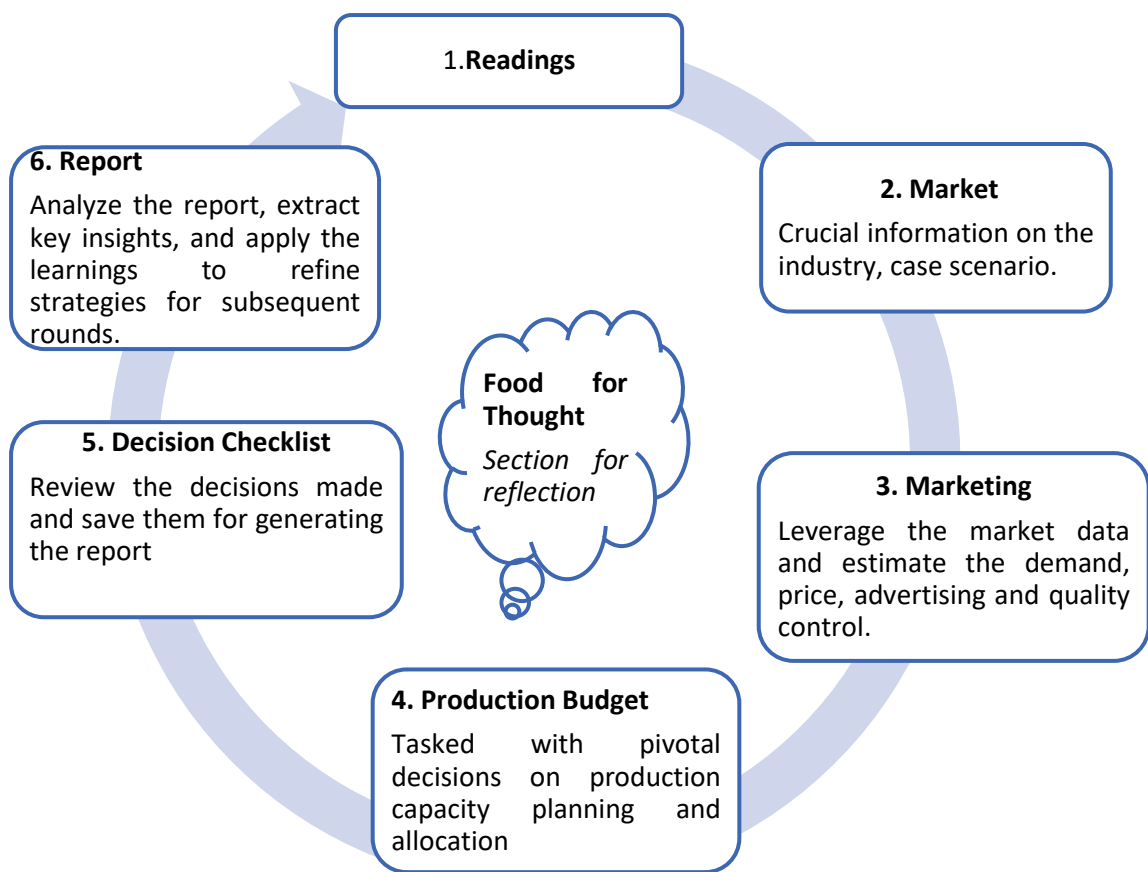


Figure 4 The Sequence of Decision-Making Process

2. Game Arena

2.1 Introduction

This page encompasses information detailing the functioning of the simulation and provides insights into each tab's decision-making.

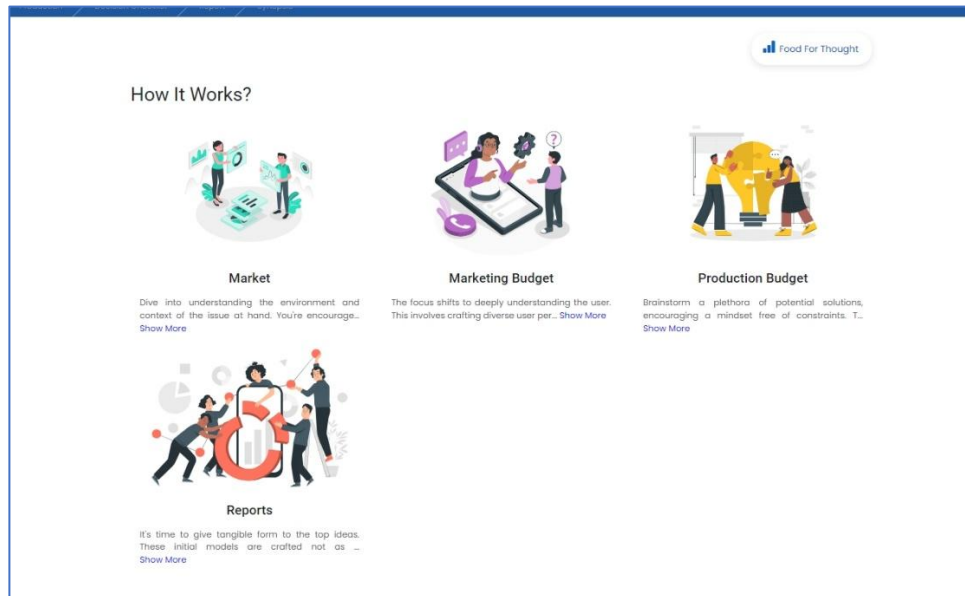


Figure 5 The Introduction Tab

2.2 Food for Thought

This segment provides crucial aspects to consider, offering valuable insights for decision-making.

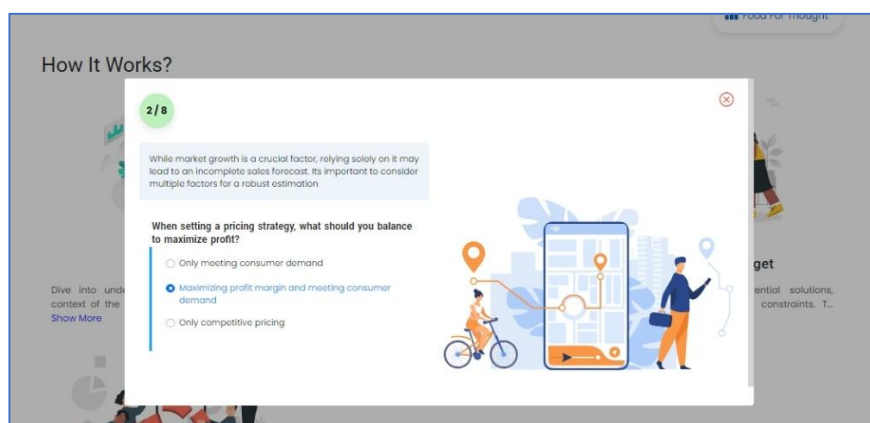


Figure 6 The food for Thought Section

2.3 Market

Prior to commencing the decision-making process, it is essential to carefully examine the "Market" section. This section offers vital information on case scenario, industry overview and the key parameters to consider when making decisions in each tab.

Summary of the Market Section:

In a dynamic manufacturing facility specializing in high-quality products, the company holds a significant market share. The local market exhibits steady growth, presenting strategic opportunities for expansion. As the appointed manager, optimizing profit margins across all product lines is a key priority.

Competitor analysis is essential to staying competitive. One competitor is known for its aggressive pricing strategies, while the other two focus on delivering premium-quality products and leading in innovation with unique designs. Understanding their strengths and weaknesses helps inform strategic decisions on pricing, quality, and design to maintain a competitive edge.

Aligning with emerging trends, such as sustainable and eco-friendly practices, can improve brand perception while addressing evolving consumer demands. Concurrently, optimizing production processes and leveraging innovative technologies helps reduce costs and enhance profit margins. These combined efforts position the company for ongoing success in an evolving and competitive market.

In our dynamic clothing manufacturing facility nestled in Coimbatore, we specialize in crafting high-quality Jeans and Tops with a market share of 30%. The local market exhibits promising growth at a rate of 5%, providing a strategic opportunity for our company to expand its market presence. As the appointed manager of our factory, participants will play a pivotal role in optimizing profit margins for both product lines. Situated advantageously in Coimbatore, we anticipate a surge in orders driven by our commitment to competitive pricing and uncompromising product quality.

Navigating the local market involves a keen understanding of our three major competitors. DenimCraft, known for its aggressive pricing strategies, poses a significant market presence. LuxeThreads, a key competitor, focuses on delivering premium-quality products, attracting a discerning customer base. Meanwhile, FashionInnovate leads in innovative designs, carving out a distinctive niche. Recognizing the strengths and weaknesses of these competitors empowers us to make strategic decisions on pricing, quality, and design, ensuring we maintain our competitive edge.

Our commitment to staying ahead involves aligning with recent market trends, notably the growing preference for sustainable and eco-friendly fashion. By incorporating environmentally conscious practices into our production processes, we not only meet evolving consumer expectations but also contribute to a positive brand image. Concurrently, optimizing our production efficiency is imperative for achieving a cost advantage. Exploring innovative technologies and refining manufacturing processes allows us to reduce production costs, thereby enhancing profit margins for both our Jeans and Tops. With an eye on market dynamics and a focus on production optimization, we position our factory for continued success in this competitive and ever-evolving market.

Parameters

	Jeans	Top
Sales, units	120000	180000
Price per unit	800	600
Variable cost per unit	500	350
Contribution per unit	300	250
	Jeans	Top
Previous Period Financials, mn INR		
Revenue	96	108
Variable cost	60	63
Gross profit	36	45
Advertising	5	7
Quality control	8	6
Administration cost	10	11
Company-wide		
Production		
Production line cost, mn INR		120
Depreciation, years		10
Production Capacity, units		310000
Capacity reduction, %		20%
Inventory holding cost per unit, INR		90

Figure 7 The Market Tab

2.4 Marketing

The decision-making process begins in the Marketing section, where participants have to estimate product demand, set prices, and allocate budgets for advertising and quality control. By leveraging market data, parameters, and demand trends, participants can make informed decisions. Setting competitive and attractive prices is essential for maximizing sales and profitability, while taking market demand and cost structures into account. Budgeting for promotional activities is crucial to enhance product visibility, attract potential customers, and

drive sales. Investing in quality control processes ensures that products meet high standards, thereby boosting customer satisfaction and strengthening brand reputation.

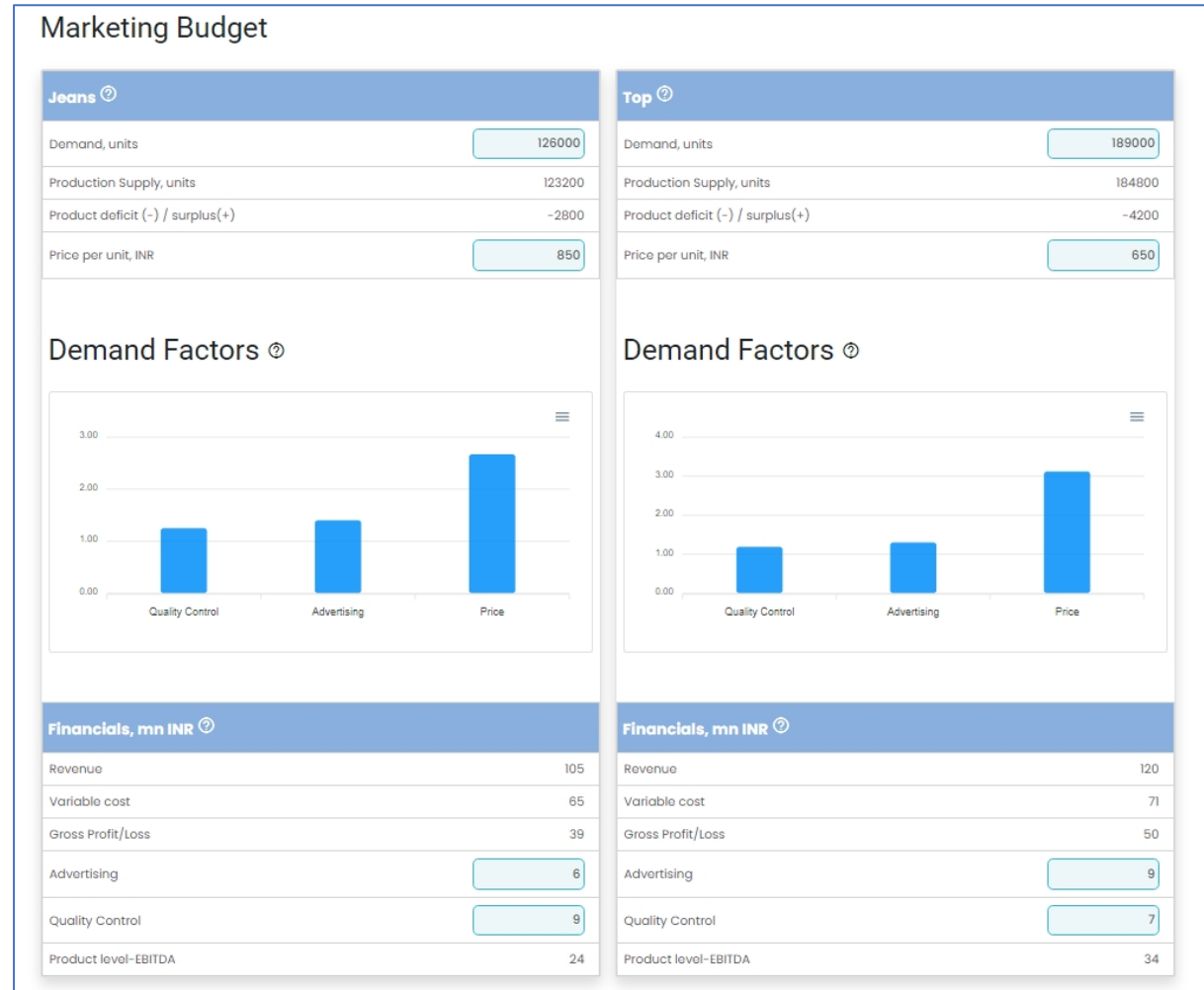


Figure 8 Marketing decision area in the game Arena

2.5 Production

On the Production page, participants manage production planning and capacity allocation. Their objective is to synchronize production with projected demand by potentially investing in plant expansion to boost capacity. Capacity allocation involves efficiently distributing

production resources across different products or lines, aiming to meet demand without overwhelming existing facilities.

Planning ②		Investments ②	
Production Capacity, units	248000	Number of Machinery	4
New capacity, units	60000	Machinery Cost, mn INR	60
Total Production Capacity, units	308000		
Production Planning, units	308000		
Capacity Utilization, %	100		

Jeans Planning ②		Top Planning ②	
Capacity Allocation %	40	Capacity Allocation %	60
Variable cost per unit, INR	530	Variable cost per unit, INR	382
Production, units	123200	Production, units	184800
Inventory, units	0	Inventory, units	0
Inventory, mn INR	0.0	Inventory, mn INR	0.0

Figure 9 The planning area under the Production Tab in Game Arena

2.6 Decision checklist

The Decision Checklist helps participants review and adjust their decisions before the deadline. Once decisions are finalized, participants can submit them using the Submit option if they choose to proceed with the Decision Checklist. The Decision Checklist helps participants review and finalize their decisions. Once decisions are confirmed, they can submit them using the "Submit" option and choose [save and proceed](#). It's important to remember that once decisions are submitted and saved, they cannot be changed for that round. This ensures that decisions are carefully considered before finalization, promoting clarity and accountability in the decision-making process.

Decision Checklist

Round 1

User

Marketing Budget

Jeans Demand, units	126000
Jeans Price, INR	850
Jeans Advertising, mn INR	6
Jeans Quality Control, mn INR	9
Top Demand, units	189000
Top Price, INR	650
Top Advertising, mn INR	9
Top Quality Control, mn INR	7

Production Budget

Number of Machinery	4
Jeans Capacity Allocation	40%
Top Capacity Allocation	60%

Submit

Marketing

Production

Help

"You are submitting your decisions as the final decision. Do you want to save and proceed with reports?"

Close

Save & Proceed

Decision Checklist

Round 1

User

Marketing Budget

Jeans Demand, units	126000
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Figure 10 The Decision Checklist

2.7 Report

The report section provides the details on the participants' critical thinking abilities in each aspect. It demonstrates how their thought process influenced the outcomes, providing insights into the impact of their decision-making skills.

This section consists of feedback on the performance of the participant based on four metrics.

- **Rigour:** Consistency throughout the decision-making and recommendation.
- **Structuring:** Logical approach of gathering and analysing information
- **Synthesis:** Approach towards interpreting information and making informed decisions
- **Business Judgement:** Ability to evaluate facts, and consider risks and possible implications when making decisions.

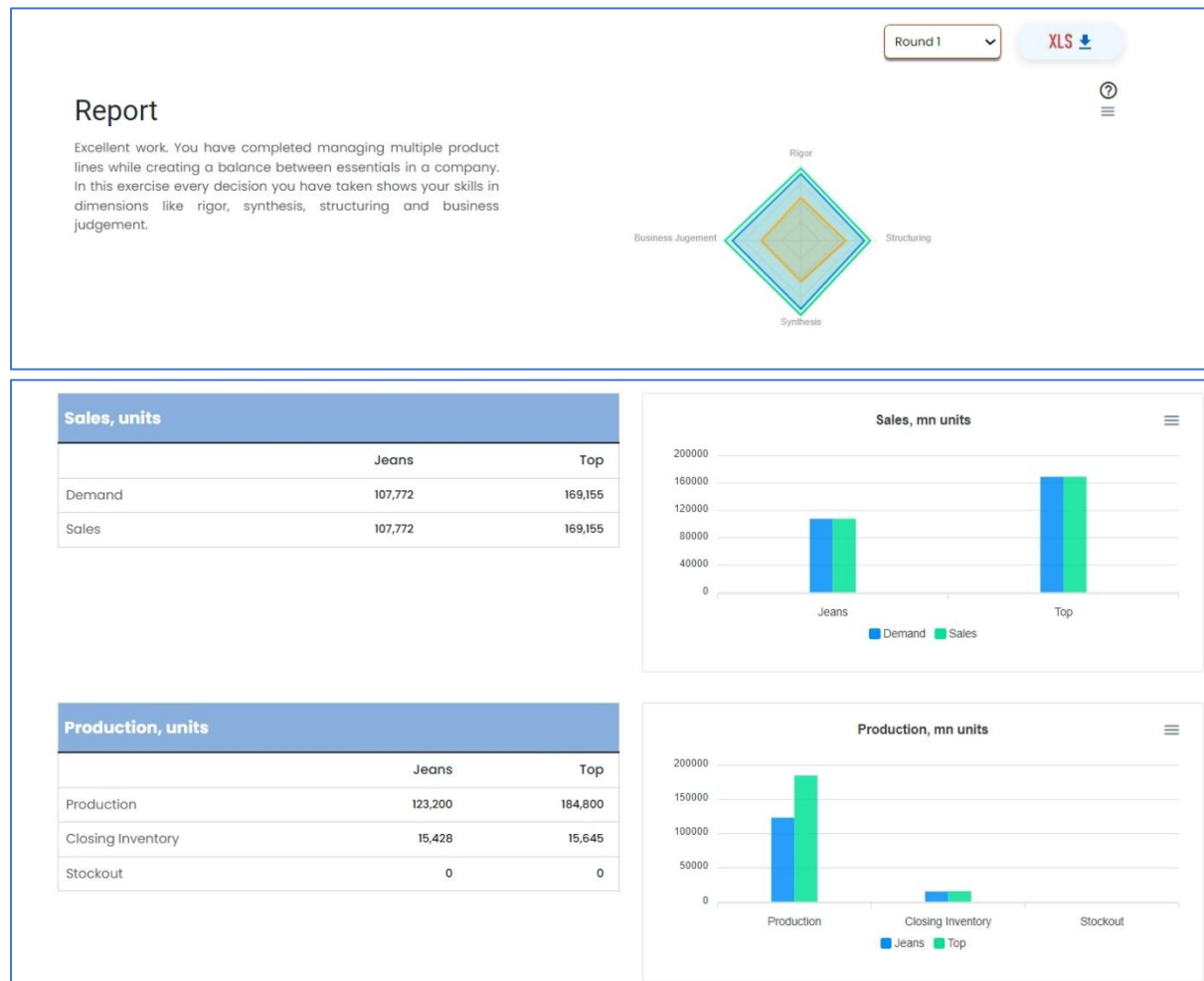


Figure 11 The Report Section

In the report section, the participants can see the sales achieved vs the demand set for the round. The details on the production planning, whether there is any stockout or not. The report also consists of the operating income, KPI product level such as the Operating margin, contribution Ratio, market share. The break-even point contains information on the break-even sales and margin of safety. KPI company level includes company level operating margin, market share, operating profit or loss.

Sales Achieved vs. Demand Set: This section compares the actual sales figures to the forecasted demand for the round. A higher actual sales figure compared to the demand set indicates better-than-expected performance, while lower sales suggest a shortfall in meeting market expectations.

Production Planning Details: This part provides insights into how well the production was planned. It includes information on whether there was a stockout (a situation where demand exceeds supply, leading to missed sales opportunities) or if production met or exceeded the demand.

Operating Income, mn INR		
	Jeans	Top
Revenue	92	110
Production cost	57	65
Gross Profit/Loss	34	45
Advertising	6	9
Quality	9	7
Inventory holding cost	1	1
Administration	9	11
EBITDA	9	17
Depreciation	7	11
Operating Profit/Loss	2	6

Break even point		
	Jeans	Top
Breakeven sales, mn INR	85	89
Breakeven sales, units	99,953	137,299
Margin of safety, units	7,819	31,856

KPI company-wide	
Operating Margin	4%
Market Share	26%
Operating Profit/Loss, mn INR	8

KPI product level		
	Jeans	Top
Contribution Ratio	26%	30%
Operating Margin	2%	6%
Market Share	26%	27%

Figure 12 The Report Section

Operating Income: Operating income is the profit generated from the core business operations, excluding any income from investments or taxes. It is a crucial measure of the company's efficiency in managing its primary business activities.

KPI Product Level:

- **Contribution Ratio:** This metric shows the portion of sales revenue that contributes to covering fixed costs and generating profit. It is calculated as $(\text{Sales} - \text{Variable Costs}) / \text{Sales}$.

- **Operating Margin:** This ratio measures the percentage of revenue that remains after deducting operating expenses. It indicates the profitability of the product before interest and taxes.

- **Market Share:** Market share represents the percentage of total sales in the market attributed to a specific product. It provides insights into the product's competitive position in the market.

Break-even Point:

- **Break-even Sales:** The level of sales at which total revenues equal total costs, resulting in neither profit nor loss. It helps in understanding the minimum sales needed to cover all costs.

- **Margin of Safety:** This indicates how much sales can drop before the company reaches its break-even point. A higher margin of safety suggests a more secure position against sales fluctuations.

KPI Company Level:

- **Company-Level Operating Margin:** Similar to the product-level operating margin, but calculated for the entire company. It shows the overall profitability of the company's operations.

- **Company-Level Market Share:** The total market share held by the company across all its products. It indicates the company's overall presence and competitiveness in the market.

- **Operating Profit or Loss:** The net result of the company's operations after deducting operating expenses from operating income. A positive value indicates profit, while a negative value indicates a loss.

2.8 Synopsis

The synopsis section offers a concise summary of key decision areas and outcomes, highlighting critical insights including the KPI achieved and strategic directions taken during the simulation, facilitating comprehensive understanding and future planning.

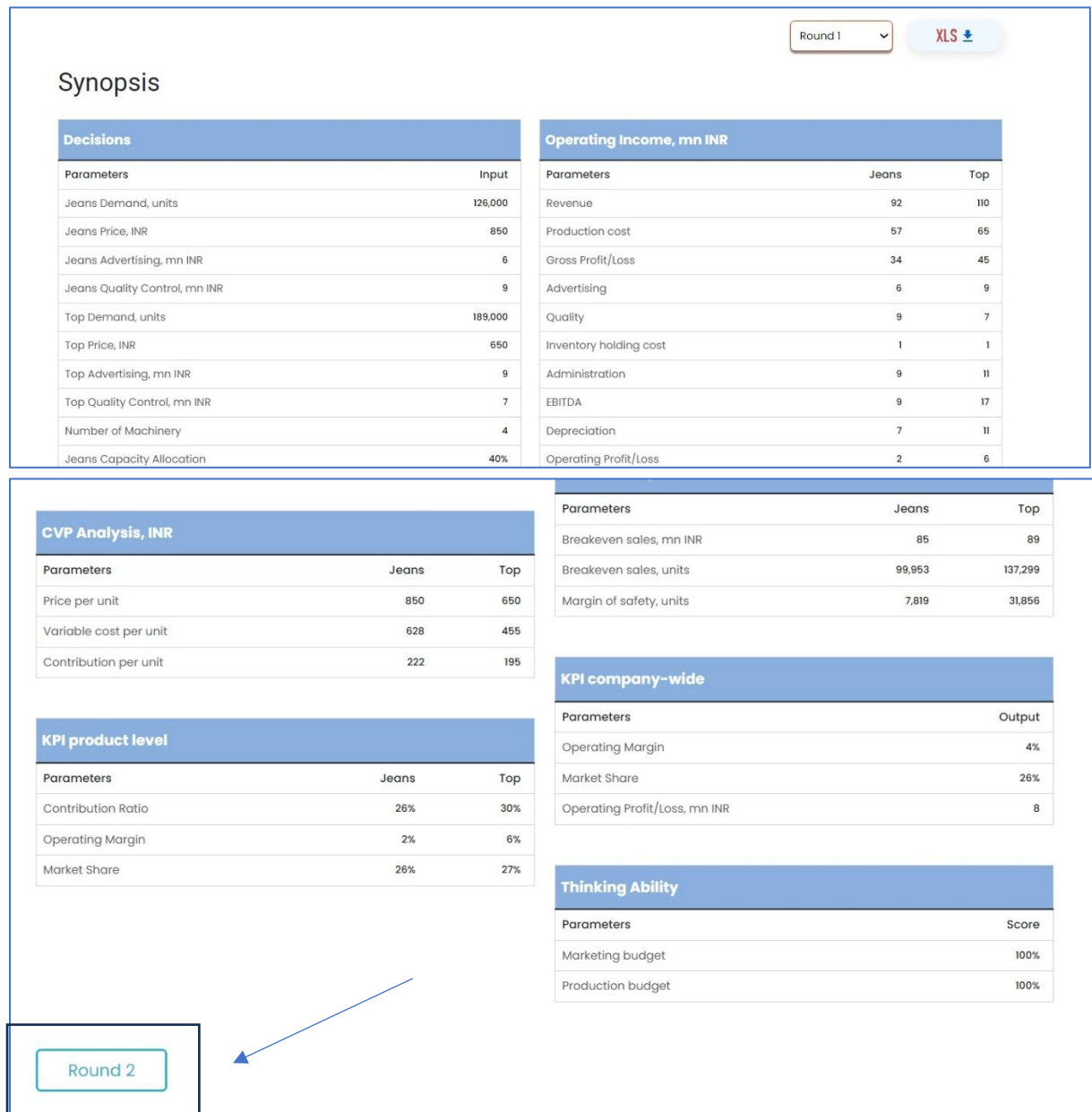


Figure 13 The Synopsis

To proceed to the next round, click on the “next round” at the bottom of the synopsis section. This will redirect participants to the home page, where they can access the next round by clicking on the corresponding card.

2.9 One of the Sample Ideal Solution

It is essential to acknowledge that numerous combinations can lead participants to achieve the desired outcomes, and the provided solution is just one example for faculty reference and not intended for direct communication to students.

Decision Area	Marketing Budget	Input
Jeans	Demand, Units	107,000
	Price, INR	1,050
	Advertising, mn INR	20
	Quality Control, mn INR	15
Tops	Demand, Units	170,000
	Price, INR	700
	Advertising, mn INR	10
	Quality Control, mn INR	11
Production Budget	Number of Machinery	4
Capacity Allocation	Jeans	37%
	Top	55%

3. Implementation – Methodologies

From our experience, we have seen Professors using two different teaching plan options. Option A is for the one who is pressed on time while option B offers a chance to participants to use the module multiple times and learn from their past decisions.

Option A

To begin, the instructor should introduce the module with the walkthrough video, explaining to the students all the different decision-making points involved in the module and how to connect them with one another (with the use of teaching notes provided by Cesim). The instructor should then provide the opportunity for the students to play the module by themselves. The students can play the module within a team or individually. Once the module is over, the instructor can quickly review the decision-making made by the students along with the results they achieved. The instructor can compare the results of different students across the module and then debrief them while driving a discussion in the class.

1. Introduction (5 minutes)
2. Run Simulation (30 to 35 minutes)
3. Debrief (10 minutes)
4. Conclusion (8 to 10 minutes)

Option B

To begin, the instructor should introduce the module with the walkthrough video, explaining to the students all the different decision-making points involved in the module and how to connect them with one another (with the use of teaching notes provided by Cesim). The instructor should then provide the opportunity for the students to play the module by themselves. The students can play the module within a team or individually. The instructor should debrief them in short. The instructor should ask the students to play the module again with the same scenario or if they want to change some parameters that can be done. The students again played the module and completed it.

Once the module is complete the instructor can quickly review the decision-making made by the students along with the results, they achieved across the rounds to compare their progress, trends, and approach towards the module. The emphasis of this plan is to let the students play the module multiple times so they can learn from their previous rounds.

The instructors generally ask students to play the module 3 times, whereas the 3rd time is for inculcating concepts.

1. Introduction (5 minutes)
2. Run Simulation (30 to 35 minutes)
3. Debrief (10 minutes)
4. Run Simulation (30 to 35 minutes)
5. Conclusion (8 to 10 minutes)
6. Run Simulation (self-paced at their own time)
7. Short Presentation & Takeaway.

3.1 Concepts that can be Covered

Here is the summary of key concepts that can discussed through the Microsimulation.

Concept	Explanation	Scenario	Action
Break-even Sales	Revenue equals total costs, no profit or loss.	Fixed costs Rs100,000, contribution margin Rs20/unit, break-even at 5,000 units.	Run promotions or adjust pricing to hit break-even sales.
Margin of Safety	Cushion between actual sales and break-even sales.	Sales at 6,000 units, break-even at 5,000; margin of safety is 1,000 units	Cut costs or boost sales to increase margin of safety.
Contribution Margin	Revenue left after covering variable costs.	Rs50 selling price, Rs30 variable cost, Rs20 contribution margin.	Raise prices or lower variable costs to improve contribution margin.

CVP Relationships	Interaction between cost, volume, and profit.	Price drops from Rs50 to Rs40; higher sales needed to cover reduced margin.	Model outcomes before making pricing adjustments.
Break-even as Strategic Tool	Helps set sales targets and manage risk.	Determine units needed to break even on a new product launch.	Align sales/marketing initiatives to meet break-even quickly.
Margin of Safety for Risk Mgmt.	Higher margin of safety indicates more stability.	30% margin of safety vs. 5% during an economic downturn.	Boost sales or reduce fixed costs to improve safety.
Pricing Strategy & Break-even	Price changes affect contribution margin and break-even sales.	Raising prices by 10% reduces break-even sales.	Analyze break-even impact before price changes.
Sales Volume vs. Contribution	Balancing higher sales with better margins.	Bulk T-shirt sales (low margin) vs. high-margin designer jeans.	Determine optimal strategy for profit using sales forecasts.
Stockout vs. Break-even	Stockouts lead to missed sales, hurting break-even targets.	Stock 4,500 units when 5,000 are needed for break-even, causing losses.	Optimize inventory levels to avoid stockouts.
CVP Sensitivity Analysis	Small changes in costs, volume, or price affect break-even.	Rs5 increase in variable costs raises break-even from 5,000 to 6,667 units.	Identify risks and adjust costs or pricing accordingly.
Scenario Planning	Simulate best/worst-case scenarios for break-even.	Base: Rs50 price, Optimistic: Rs60 price, Pessimistic: 20% cost increase.	Create contingency plans based on scenarios.
Improving Operating Margin	Actions to boost revenue or reduce costs using CVP insights.	Lower supplier prices, raise premium item prices, focus on selling high-margin products.	Regularly review CVP data to exceed break-even and improve profitability.

3.2 Case Customization

The case can be customized based on the faculty's requirements, providing the flexibility to enable or disable specific decision tabs within each module. Additionally, various parameters, such as market, financial statements, etc., can be adjusted to meet specific needs.

3.3 Suggested Discussion Starters

1. How can break-even analysis guide new product launches?
2. What steps help mitigate rising variable costs increasing break-even?
3. How can businesses balance price increases and cost cuts for better margins?
4. How can CVP scenario planning help manage market uncertainties?
5. How does a price change impact break-even, margins, and profits?
6. Why is a high margin of safety critical for managing sales volatility?
7. How does product mix affect a company's ability to meet break-even?
8. Should businesses prioritize high volume or high-margin sales?
9. How do stockouts impact meeting fixed-cost coverage?
10. When should companies shift fixed costs to variable costs?
11. How does sensitivity analysis improve pricing and cost decisions?
12. How do discounts impact break-even sales and profit?
13. What CVP strategies help improve operating margins?
14. How often should businesses track break-even KPIs for decision-making?
15. Why is break-even analysis essential for setting sales targets?
16. How do rising fixed costs impact break-even and risk?
17. What are the risks of a high break-even point in competitive markets?
18. Can break-even analysis assess sales and marketing performance?
19. How can break-even adjust for seasonal sales shifts?
20. How can automation reduce variable costs and lower break-even?